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INVERTEBRATE DRIFT AND FEEDING HABITS OF JUVENILE CHINOOK SALMON IN THE UPPER SACRAMENTO RIVER, CALIFORNIA

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The diurnal feeding habits of juvenile chinook salmon, *Oncorhynchus tshawytscha*, rearing in the upper Sacramento River from April to June 1996 were examined in relation to food availability. Daily mean drift densities in the Sacramento River ranged between 211 and 2,100 organisms/100 m³. Aquatic taxa, especially Chironomidae (Diptera) and Baetidae (Ephemeroptera), dominated drift samples and salmon stomach contents. Selection of prey by juvenile chinook salmon appeared to be based on size, abundance, or conspicuousness of organisms, as indicated by linear selection indices. The gape width of juvenile salmon increased predictably as the fish grew in length, but the sizes of prey consumed fell considerably short of gape potential due to the relative rarity of large prey in the environment. Mean stomach fullness was 2.4% of fish weight, indicating reasonable feeding opportunity during spring of wet years similar to 1996.

INTRODUCTION

Juvenile chinook salmon, *Oncorhynchus tshawytscha*, generally spend from several weeks to several years rearing in freshwater before migrating to the ocean. Most juvenile chinook salmon in the Sacramento River display an ocean-type life history strategy (Healey 1983), emigrating as subyearling fry or smolts. Although the time spent in the upper river can be short, growth in size and stored energy during early rearing can ultimately determine survival to the estuarine or ocean phase (Higgs et al. 1995). Consideration of important features of the environment, such as food availability, along with the response of the fish (e.g., food selection) to those habitat features, allows for a greater understanding of the role of freshwater rearing in the life history of juvenile chinook salmon.

The feeding habits of juvenile chinook salmon in other river systems have been studied extensively (Becker 1970, 1973; Sagar and Glova 1987, 1988; Rondorf et al. 1990). Schaffter et al.² (1982) described the drift and diet of juvenile chinook salmon

¹ Current address: 3059-E National Fish Hatchery Road, Hagerman, Idaho 83332.

² Schaffter, R.G., P.A. Jones, and J.G. Karlton. 1982. Sacramento River and tributaries bank protection and erosion control investigation: Evaluation of impacts on fisheries. Final report. California Department of Fish and Game, Red Bluff, California, USA.

in the Sacramento River in 1981, and Merz and Vanicek (1996) did the same for the lower American River, a Sacramento River tributary, in 1992 and 1993; however, recent published studies focusing on food availability and feeding habits of juveniles in the upper mainstem Sacramento River are lacking. Fall-run, late-fall-run, winter-run, and spring-run chinook salmon stocks use the upper Sacramento River and its tributaries for spawning. Numerous anthropogenic changes to the river, such as construction of dams, flow regulation and diversion, stream channelization, conversion of vast areas of floodplain to agriculture, and gravel extraction, among others, have contributed to the decline of all 4 stocks (Brown 1991, Fisher 1994). The listings of the winter run as endangered (NMFS³ 1994) and the spring run as threatened (NMFS⁴ 1999) under the federal Endangered Species Act limit the sacrifice of specimens for diet studies, whereas, ironically, the stocks' imperiled status makes understanding their life history, behavior, and habitat requirements ever more crucial. Given the possibility of future endangered species listings of the other Sacramento River chinook salmon stocks, it is important for conservation purposes to understand the role that environmental factors play in limiting salmon populations. The present study provides insight into seasonal food availability and diet that is applicable to all stocks of salmon in the river, though only fall-run young-of-the-year were sacrificed for stomach analyses.

The main objectives of this study were to describe 1) the availability of natural foods (invertebrate drift) in the upper mainstem of the Sacramento River during April–June 1996, 2) the taxonomic composition of organisms in the diet of juvenile chinook salmon rearing in the upper mainstem river during that time, 3) prey selection by juvenile salmon, 4) size characteristics of prey in the diet of juvenile salmon in relation to fish size, and 5) feeding intensity and energy intake of juvenile salmon.

METHODS

Study Area

Originating near Mt. Shasta in northern California, the Sacramento River extends approximately 595 river km (rkm) along the floor of the Central Valley to its mouth in the Sacramento-San Joaquin Delta. Bordered by the Sierra Nevada and Cascade ranges on the east and the Klamath and Coast ranges on the west, the northern Central Valley is characterized by heavy precipitation during winter and spring and negligible precipitation during summer and fall.

³ NMFS (National Marine Fisheries Service). 1994. Endangered and threatened species; status of Sacramento River winter-run chinook salmon. Federal Register 59:440-450.

⁴ NMFS (National Marine Fisheries Service). 1999. Endangered and threatened species; threatened status for two chinook salmon evolutionarily significant units (ESUs) in California; final rule. Federal Register 64:50393-50415.

Fed primarily by runoff from precipitation and snowmelt from surrounding mountains, the Sacramento River at Red Bluff historically exhibited flow ranging from about 28 m³/s in summer to 100-year flood levels of about 11,900 m³/s (CDWR⁵ 1994a). River discharge is now regulated by the Shasta-Keswick Dam complex (rkm 501 and rkm 486), the operation of which tends to dampen oscillation in the hydrograph, as compared to historic flows (CDWR⁶ 1994b). River flow in the study area largely depends upon the management of water projects upstream. Major water diversions located in the study area include the Anderson-Cottonwood Irrigation Diversion (rkm 481), the Red Bluff Diversion Dam (RBDD) (rkm 391), and the Glenn-Colusa Irrigation Diversion (rkm 332). The river provides the main water supply for the cities of Redding, Chico, and Red Bluff, in addition to supporting agricultural and recreational water uses.

Study Sites and Duration

We investigated the available foods for juvenile chinook salmon at 5 sites on the upper mainstem Sacramento River between rkm 380 and rkm 430 during April–June 1996 (Table 1, Fig. 1). Feeding habits data from juvenile fall-run chinook salmon were collected at 9 sites between rkm 311 and rkm 444 (Table 1, Fig. 1) during the same period.

Drift Collection and Handling

Field Collections of Invertebrate Drift

Food abundance was estimated by arranging up to 4 replicated drift nets in a transect perpendicular to the river's edge. Sampling was repeated throughout the day at a given site. The drift samples were collected between sunrise and sunset to compare with daytime feeding habits data. Drift was sampled at 1 site per week, not concurrently with fish sampling efforts.

Two types of drift nets were used. The 1st type had a mesh size of 260 μ m and mouth dimensions of 45 x 30 cm. These nets could be oriented with either the long or short side parallel to the substrate and were most appropriate for sampling shallow near-shore areas. The second net type had a 75 x 15.5 cm mouth and 64 μ m mesh, and was most often used in deeper water farther from shore, where the shorter nets could not capture surface drift. Drift nets sampled the water for 15–25 minutes. After removal of a net from the water, the contents were washed down to the closed end of the net. The sample was sieved to remove excess water and preserved in 10% formalin. Rose bengal dye was added to each sample to facilitate sorting of organisms.

⁵ CDWR. 1994a. Sacramento River bank erosion investigation progress report. California Department of Water Resources, Northern District, Red Bluff, California, USA.

⁶ CDWR. 1994b. Use of alternative gravel sources for fishery restoration and riparian habitat enhancement in Shasta and Tehama counties, California. California Department of Water Resources, Northern District, Red Bluff, California, USA.

Table 1. Sites and dates of fish and drift collections in the upper Sacramento River, April–June 1996.

Site (rkm)	Fish collection dates	Drift collection dates
444	4/26	not sampled
438	4/12, 5/10, 6/21	not sampled
430	not sampled	4/7, 5/19, 6/30
415	4/12, 4/26	4/14, 6/2
396	4/11, 4/26, 5/9	5/3, 6/23
391	4/20, 5/23	4/20, 5/13, 6/23
390	4/25, 5/23	not sampled
380	4/11, 4/25, 5/9, 5/23, 6/6, 6/20	5/5, 6/9
352	4/11, 4/25, 5/9, 5/23, 6/6, 6/20	not sampled
311	4/11, 4/25, 5/9, 5/23	not sampled

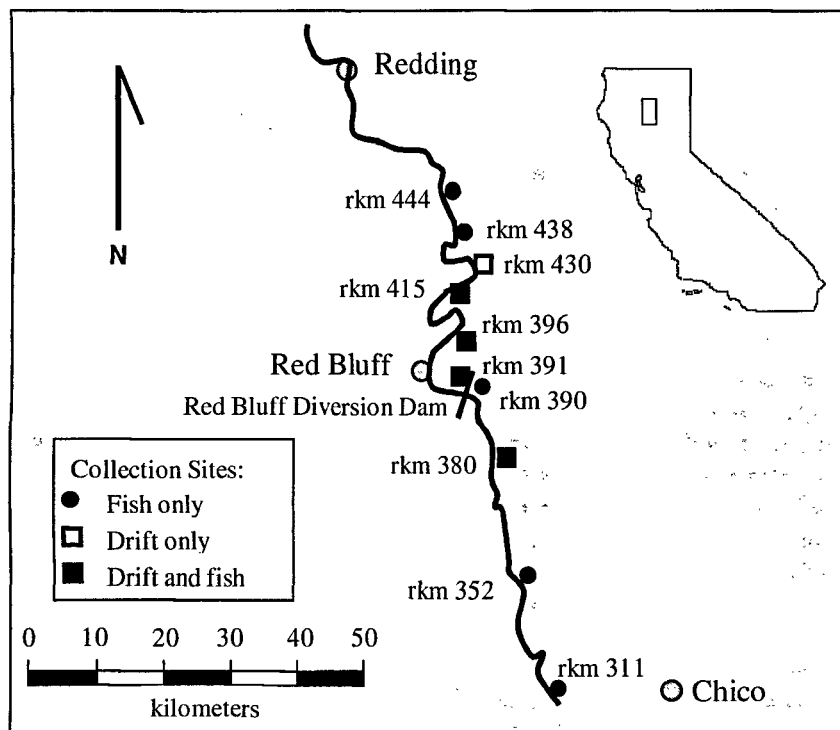


Figure 1. Map of collection sites for invertebrate drift and juvenile chinook salmon in the upper Sacramento River, April–June 1996.

Drift Sample Processing

To expedite processing, some drift samples were suspended in water and divided into subsamples using a Folsom plankton splitter. In order to identify a minimum of 100 organisms per sample, sorting ranged from an entire sample to as little as 1/8 of

a sample. The sample or subsample was viewed under a lighted magnifying lens. Invertebrates and fish larvae were sorted from the sample and transferred to a bottle of 80% ethanol. Sorted organisms were identified using published keys (Usinger 1956, Pennak 1978, Merritt and Cummins 1984, Arnett 1985). Drift density (DD, the number of organisms per 100 m³ of water) was estimated with the equation:

$$DD = 100 \times (\text{number of organisms per net-hour} / \text{m}^3 \text{ filtered per net hour}).$$

The number of organisms per net-hour was calculated after adjusting for subsample size.

Juvenile Salmon Collection and Handling

Fish Collection

Beach seines (1.21 m × 9.09–22.73 m, 3.2-mm stretch mesh) were used to collect juvenile chinook salmon from gravel bar and boat ramp sites. Sites were sampled between 0800 and 1600 hours once every 2 weeks. After capture, juvenile salmon were anesthetized with tricaine methanesulfonate (MS-222), and each individual was assigned to a run (fall, late-fall, winter, or spring) based on fish length and date of capture using a run identification model (Fisher⁷ 1992, as modified by Greene⁸ 1992). A subsample of up to 10 fall-run salmon per site was sacrificed by prolonged exposure to the anesthetic and preserved in 10% formalin.

Diet Analysis

Juvenile salmon were examined to determine the number, size, taxonomic composition, and total wet weight of organisms in stomach contents. Preserved fish specimens were measured to the nearest millimeter (FL), blotted, and weighed to the nearest milligram. Dial calipers were used to measure the gape width of each fish specimen to the nearest 0.05 mm. The stomach was excised and blotted, and the stomach weight, with and without the contents, was determined to the nearest 0.01 mg. The total wet weight of the contents was calculated by subtraction. The stomach fullness index (SFI) (Hyslop 1980) for each fish was calculated with the equation:

$$SFI (\%) = 100 \times (\text{wet weight of stomach contents}) / (\text{fish wet weight}).$$

Stomach contents were viewed under a microscope fitted with an ocular micrometer. Each organism was identified to the lowest practical taxonomic level

⁷Fisher, F.W. 1992. Chinook salmon *Oncorhynchus tshawytscha* growth and occurrence in the Sacramento-San Joaquin River system. California Department of Fish and Game, Inland Fisheries Division, Red Bluff, California, USA.

⁸Greene, S. 1992. Estimated winter-run chinook salmon salvage at the State Water Project and Central Valley Project delta pumping facilities. Memorandum to R. Brown, 5/8/92. California Department of Water Resources, Sacramento, California, USA.

(family in most cases) and its life stage (larva/nymph, pupa, or adult) was recorded when possible. The first 10 representatives encountered from each invertebrate taxon were measured for head capsule width (HCW) to the nearest 0.03 mm, for organisms with a head capsule. Head capsule widths were used to represent relative invertebrate prey size, as fragmentation of prey during swallowing and digestion precluded the use of other measures, such as prey length. Total lengths of fish larvae found in the stomach were measured to the nearest millimeter if the condition of the specimen permitted. Copepods, cladocerans, nematodes, and aquatic worms were counted if present, but were not measured. Fragments of organisms without a head attached (legs, wings, or abdomens) were neither counted nor identified; it was assumed that the source organism had already been represented in the count of heads or head capsules.

Selection of individual taxa by juvenile chinook salmon was calculated from abundance in the drift, totaled across all sites and sampling dates, and compared with the abundance in the diet of all fish examined, using the linear selection index (L) (Strauss 1979). The index is described by the equation:

$$L = r_i - p_i,$$

where r_i is the relative abundance (expressed as a proportion) of taxon i in the diet of chinook salmon and p_i is the proportion of taxon i in the drift during the same time period. The index ranges from +1 to -1, with positive values indicating preference, negative values indicating avoidance or inaccessibility, and zero values indicating random feeding (Strauss 1979).

Statistical Methods

Student's t -tests were used to analyze the null hypothesis that $L = 0$ for each taxon (Strauss 1979). Taxonomic groups present in low numbers in both the environment and the diet were excluded from the analysis of L . Least squares regression was used to evaluate the relation between fork length and fish gape width or prey head capsule width (HCW) (mean, maximum, and minimum). For fork length-HCW regressions, only fish with >10 items in the stomach were used, to reduce variability due to low feeding intensity and low selection. To normalize the data on number of prey per stomach, the data were square-root transformed for calculation of means and confidence intervals, then reverse transformed for presentation. The significance level was $\alpha = 0.05$.

RESULTS

Drift Density

Daily mean drift density in spring 1996 ranged from 211 to 2,100 organisms/100 m³ (Table 2) and the overall mean was about 617 organisms/100 m³. Mean drift density fell below approximately 800 organisms/100 m³ on most sample dates, but 2 higher

estimates (1,537 and 2,100 organisms/100 m³) were recorded on 23 June 1996 at rkm 396 and rkm 391 (Table 2). The lowest mean drift densities were estimated for rkm 415 during April and June (211 organisms/100 m³ for both months). Aside from the 2 high estimates, daily mean drift density showed no distinct spatial patterns (Fig. 2).

Relative Abundances of Drift and Diet Taxa

The drift during April–June 1996 was dominated by aquatic taxa, which made up about 96% of organisms captured (Appendix A). The most prevalent members of the drift were larvae, pupae, and adults of the family Chironomidae (Diptera), which made up 54% of the total number of organisms (Appendix A). The high abundance of chironomids accounted for the high general contribution of the order Diptera. The family Baetidae (Ephemeroptera), particularly nymphs, also were present in large numbers in the drift, with 12% of the total. Taxa contributing 3–6% of the drift included simuliid (Diptera) larvae (4.16%), oligochaetes (5.72%), cladocerans (4.79%), copepods (4.98%), and fish larvae (3.42%, including Catostomidae). Other taxa contributed relatively small numbers to the drift.

The diets of 153 juvenile chinook salmon were examined; among these, no empty stomachs were present. As for the drift, the diet was dominated by aquatic taxa, which made up 94% of consumed items (Appendix A). Chironomids of all life stages were the predominant prey, collectively making up >63% of the diet during April–June 1996. The order Diptera made up about 68% of the diet, mostly due to the high contribution of chironomids. Baetid nymphs and adults were common prey, with a combined 14% of the total. Homopterans (3.30%), trichopterans (3.92%), and fish larvae (3.30%, including catostomids) were also found in notable numbers in salmon stomachs.

Table 2. Mean, standard error (SE), and coefficient of variation (CV%) for drift density (number of organisms per 100 m³ water filtered per net) by site and date for the upper Sacramento River, April–June 1996.

Site (rkm)	Date	Number of samples	Mean drift density (No./100 m ³) (+/-SE)	CV%
430	4/7/96	6	805 (270)	82
	5/19/96	11	638 (169)	88
	6/30/96	9	295 (54)	55
415	4/14/96	11	211 (34)	54
	6/2/96	10	211 (32)	48
396	5/3/96	2	483 (161)	47
	6/23/96	2	1,537 (291)	27
391	4/20/96	6	484 (107)	54
	5/13/96	14	670 (201)	112
	6/23/96	9	2,100 (1,192)	170
380	5/5/96	15	363 (64)	69
	6/9/96	12	538 (125)	81

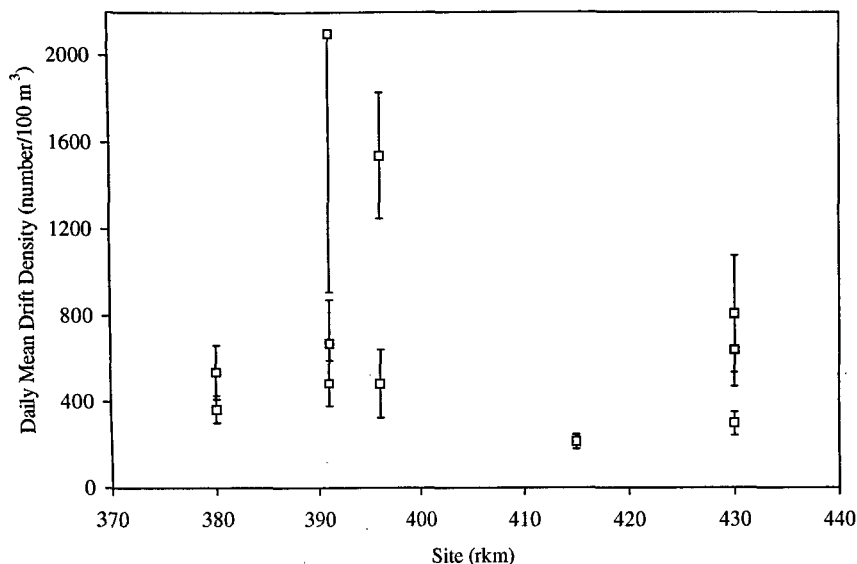


Figure 2. Daily mean drift density ($\pm$ SE) plotted as a function of site location on the Sacramento River for April–June 1996. The upper SE limit (3,292) for the highest value at rkm 391 was omitted for scaling purposes.

Selection of Available Prey by Juvenile Chinook Salmon

For many groups, the linear selection index was not significantly different from 0, suggesting random selection of those items. Values of L ranged from -0.19 to a maximum of 0.17 (Fig. 3), reflecting the conservative nature of the index, which only approaches extreme values when an item is rare but consumed almost exclusively, or is highly abundant but rarely consumed (Strauss 1979). Among the taxa numerically common in the drift, significant negative values of L were observed for chironomid larvae (-0.19), oligochaetes (-0.06), and cladocerans (-0.05), whereas significant positive values were calculated for chironomid pupae (0.12) and chironomid adults (0.17) (Fig. 3). Collembolans, baetid nymphs, aphid (Homoptera) adults, simuliid larvae, hydropsychid (Trichoptera) larvae, hymenopterans, and water mites (Hydracarina) displayed weak, but significant, selection values (Fig. 3).

Fish Size, Gape Size, and Prey Size Relations

The gape width of juvenile chinook salmon showed a significantly positive linear relation with fork length ($y = 0.0864x - 0.5658$; $r^2 = 0.82$, $n = 153$, $P < 0.0001$) (Fig. 4). The slope of the relation between fork length and head capsule width (HCW) of measurable invertebrate prey was highest for maximum HCW and lowest for minimum HCW (Fig. 4). The amount of variability in HCW explained by fork length also decreased from maximum to mean to minimum HCW. The relation between fork length and maximum HCW was significantly positive ($y = 0.0238x - 0.3109$; $r^2 = 0.32$, $n = 113$, $P < 0.0001$), as was the relation between fork length and mean

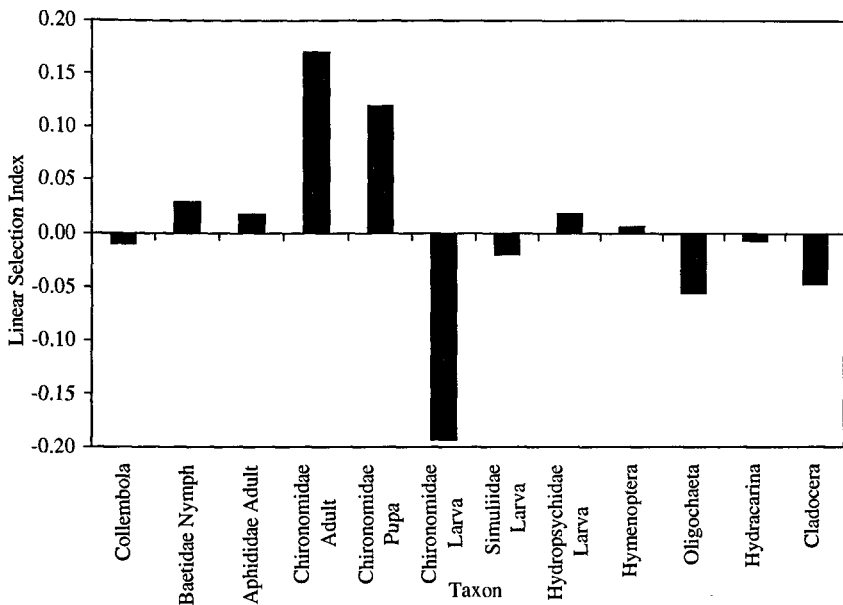


Figure 3. Significant linear selection indices (L) for common taxa in the drift and diet of juvenile chinook salmon in the upper Sacramento River, April–June 1996. Possible values of L range from -1 to 1.

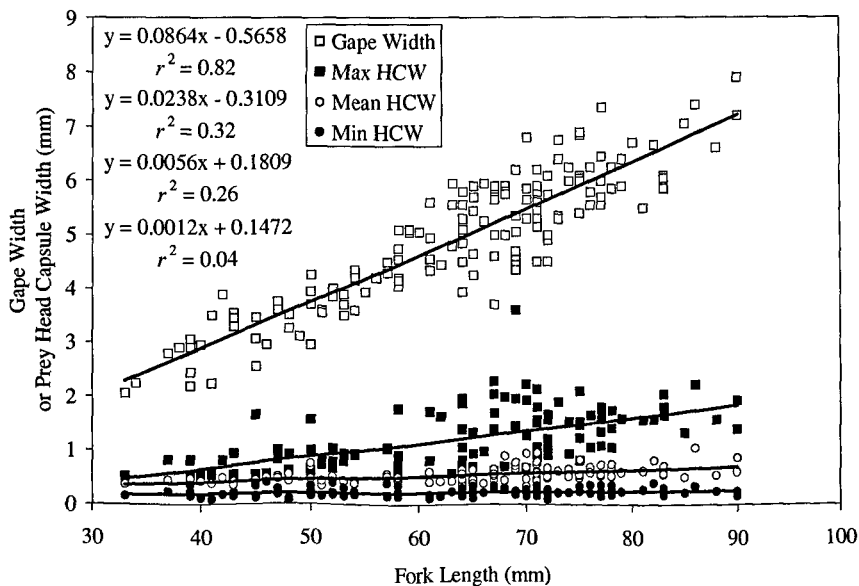


Figure 4. Regressions of gape width ($n = 153$), and maximum, mean, and minimum head capsule width ($n = 113$) of invertebrate prey on fork length of juvenile chinook salmon captured April–June 1996 from the upper Sacramento River.

HCW ($y = 0.0056x + 0.1809$; $r^2 = 0.26$, $n = 113$, $P < 0.0001$). Although the regression of minimum HCW on fork length was significantly different from 0, it explained only 4% of the variability in minimum HCW ($y = 0.0012x + 0.1472$, $r^2 = 0.04$, $n = 113$, $P < 0.05$). The mean gape width of all fish examined was 4.92 mm ($SE = 0.56$, range = 2.05–7.90 mm, $n = 153$). The mean HCW of measurable invertebrate prey items was 0.53 mm ($SE = 0.01$, range = 0.06–3.60 mm, $n = 3,086$). Chironomid larvae were the smallest of the common prey items as measured by HCW, whereas hydropsychid larvae were among the largest invertebrate prey (Table 3).

Fish larvae did not make up a large percentage of the total number of items in the diet, and the size of fish prey, measured in length, cannot be compared directly with the sizes (HCW) of invertebrate prey. However, there was a trend in the relation between the size of juvenile chinook salmon and the number of fish larvae consumed. Of the salmon ($n = 31$) that consumed fish larvae, all were ≥ 40 mm FL. Only juvenile salmon ≥ 60 mm FL had stomachs containing ≥ 5 fish larvae (the maximum was 38). Fish larvae in salmon stomachs ranged from 3 to 18 mm total length.

Feeding Intensity and Stomach Fullness

The number of prey items per stomach and stomach fullness index (SFI) were used as measures of the feeding intensity or energy intake of juvenile chinook salmon in the Sacramento River. Arrangement of mean number of prey items per stomach and the SFI into 3 size classes and 3 monthly categories revealed no clear trends in feeding intensity (Table 4). During April, the highest mean SFI was seen in the smallest size class (30–50 mm). However, in June, both the mean number of prey per stomach and the mean SFI increased with fish size. The SFI averaged about 2.4% ($SD = 1.6$, $n = 153$) and ranged from 0.5% to 8.4% of wet weight.

Table 3. Mean, standard deviation (SD), and range of head capsule widths (HCW) of common taxa in stomachs of juvenile chinook salmon captured from the upper Sacramento River, April–June 1996.

<u>Taxon</u>	<u>Life stage</u>	<u>Number measured</u>	<u>Mean HCW (+/-SD)</u>	<u>Range</u>
Baetidae	Adult	29	0.84 (0.10)	0.63–1.05
Baetidae	Nymph	325	0.74 (0.17)	0.24–1.11
Aphididae	Adult	103	0.42 (0.06)	0.27–0.60
Chironomidae	Adult	817	0.45 (0.08)	0.15–0.84
Chironomidae	Pupa	501	0.47 (0.08)	0.21–0.87
Chironomidae	Larva	585	0.27 (0.09)	0.09–0.63
Simuliidae	Larva	105	0.44 (0.11)	0.18–0.75
Hydropsychidae	Larva	118	1.13 (0.36)	0.39–2.22

Table 4. Mean number of prey per stomach and mean stomach fullness index (SFI), with 95% confidence intervals (CI), of juvenile chinook salmon captured in the upper Sacramento River, April–June 1996, arranged by month and size class. Data on number of prey were square-root transformed for calculation of means and confidence intervals, then reverse transformed for presentation.

Month	Size class (mm)	Sample size	Prey per stomach		SFI (%)	
			Mean	95% CI	Mean	95% CI
April	30–50	24	23	21–25	3.5	2.6–4.4
	51–70	36	29	27–31	2.0	1.7–2.4
	71–91	23	24	22–27	2.4	1.6–3.2
May	30–50	3	26	20–31	1.8	1.2–2.3
	51–70	31	28	25–30	2.5	1.9–3.1
	71–91	16	26	25–28	1.7	1.4–2.1
June	30–50	2	6	6–7	1.6	1.3–1.9
	51–70	3	24	17–32	1.6	1.4–1.9
	71–91	15	72	66–78	2.1	1.7–2.4

DISCUSSION

Drift Density

Drift density provides a rough index of the secondary productivity of a river. Mean drift density estimates in the Sacramento River (211–2,100 organisms/100 m³) were comparable to estimates in other rivers and were higher than estimates for systems in which food is considered limiting for drift-feeding salmonids. Bowles and Short (1988) found peak drift densities in a Texas stream to be near 1,000 organisms/100 m³ in February, 500 in May, 600 in August, and <100 in November. Drift density in the Rakaia River, New Zealand, during freshwater rearing of juvenile chinook salmon was 200–900 organisms/100 m³ (Sagar and Glova 1988), and, based on the size and condition of fish, food was not limiting. In contrast, drift densities for several southern Appalachian streams were as low as 0.29–27.77 organisms/100 m³, resulting in food limitation of resident salmonids (Cada et al. 1987).

Relative Abundances of Drift and Diet Taxa

We found the taxonomic composition of the drift and diet of salmon in the Sacramento River was similar to that described for other river systems (Becker 1973, Sagar and Glova 1987) and in a previous study of the upper Sacramento River (Schaffter et al.² 1982). As in other large rivers, drift and diet in the Sacramento River were dominated numerically by a few common taxa with a small contribution from other groups. In the Rakaia River, New Zealand, *Deleatidium* (Ephemeroptera: Leptophlebiidae) nymphs made up 85% of the diet of juvenile chinook salmon in the spring; *Deleatidium* constituted about 80% of the benthos and 48% (daytime) to 70% (nighttime) of the drift during spring (Sagar and Glova 1987). During summer, chironomids replaced *Deleatidium* as the most common prey item, though the

abundance of chironomids in the drift did not change substantially from that of spring. In the central Columbia River, juvenile chinook salmon fed mostly on chironomid adults (58–64% of the diet) and larvae (17–18%) (Becker 1973). Notonectidae (Hemiptera) and adult Hydropsychidae were also locally important in the central Columbia River, together making up around 5% of the diet of juvenile chinook salmon. During February–June 1981, the diet of juvenile chinook salmon in the upper Sacramento River (Schaffter et al.² 1982) was dominated by chironomids, baetids, and aphids, similar to what was found in our study. In the lower American River, the diet of juvenile salmon consisted primarily of chironomid pupae (50%) and baetid adults (20%) during February–July 1992, and chironomid pupae (54%) and larvae (40%) during February–July 1993 (Merz and Vanicek 1996).

Terrestrial organisms were relatively rare in the Sacramento River drift in spring 1996, constituting about 4% of the total. Estimates of the contribution of terrestrial organisms (primarily Aphididae) to the Sacramento River drift during spring 1981 varied from 0.5% to as much as 50% (Schaffter et al.² 1982). Aphids composed from 0.13–38% of the chinook salmon diet during that time. The wide contrast in importance of terrestrial organisms between the 2 studies, and in the Schaffter et al.² (1982) study, may reflect sampling during high winds or other riparian disturbances that caused aphids to become trapped in large numbers on the water's surface.

Selection of Available Prey by Juvenile Chinook Salmon

Previous studies have shown prey selection by salmon to be based on size, abundance, and visibility of prey (Higgs et al. 1995). Linear selection indices showed that relatively small organisms, such as cladocerans, collembolans, and chironomid larvae were avoided by juvenile chinook salmon in the Sacramento River. Positively selected items included chironomid pupae and adults, which are seasonally very abundant, and baetid nymphs, which are relatively large as well as abundant. Such factors as surface movements during adult emergence or conspicuous eyespots, in addition to size, may contribute to the greater visibility and, hence, positive selection of these items. Most taxa appeared to be randomly consumed ($L = 0$), or were too rare to allow calculation of L . Concurrent collection of drift and diet information, along with larger sample sizes, would be necessary to provide more reliable estimates of selectivity for rare organisms.

Note that food selection indices are not equivalent to an assessment of absolute preferences for certain taxa, but instead reflect a combination of factors including preference, prior experience, prey detectability, and prey availability. A limitation of all selection indices is that they cannot elucidate the underlying mechanism(s) of the observed patterns of selection (Strauss 1979).

Fish Size, Gape Size, and Prey Size Relationships

As expected, the gape size of juvenile chinook salmon in the Sacramento River increased predictably with fish length. Larger gapes allow salmon to potentially exploit a greater range of prey sizes as they grow. Inclusion of increasingly larger

prey in the diet would be advantageous for growth, assuming that larger prey do not cost more energetically to capture and handle. Field-Dodgson (1988) reported that the mean mouth width of emergent chinook salmon fry (2.2-mm mean gape, range 2.0–2.5 mm; mean fish size 33.5 mm FL) corresponded to the maximum HCW of *Deleatidium* nymphs found in the gut, which suggests that juvenile chinook salmon will consume the largest prey possible. In the Sacramento River, larger salmon ate larger prey on average, but the maximum invertebrate prey sizes in the diet fell considerably short of fish gape potential, as was observed for juvenile Atlantic salmon from Catamاران Brook, New Brunswick (Keeley and Grant 1997). Wainwright and Richard (1995) proposed that pharyngeal gape rather than oral gape may set the limit on the size of prey consumed, which would translate into maximum prey being about half the size of mouth gape. However, as the authors noted, distortion of prey during swallowing complicates that hypothesis. Even if pharyngeal gape determined the size of prey successfully consumed by juvenile chinook salmon in the Sacramento River, maximum HCW of prey still fell considerably short of this limit.

The weak increase of mean and minimum invertebrate prey size with increasing fish size indicates that all juvenile salmon in the upper Sacramento River included small prey items in their diet to meet energy requirements, despite the ability of larger fish to consume larger prey. Essentially, invertebrate prey size was determined more by availability in the environment than by morphological constraints. Though the ability of juvenile salmon to capture fish larvae also appeared to increase as the salmon grew in size, the actual availability of such prey would be only indirectly measurable by drift methods. The relationship between salmon size and piscivory is more complicated, as fish prey are evasive and must be located and pursued, whereas drift feeding is more a process of maintaining a stationary position suitable for intercepting drifting prey. Therefore, with regard to piscivory, gape constraints as well as efficiency in pursuit or handling of fish prey could explain the relationship between predator size and the number of prey consumed.

Feeding Intensity and Stomach Fullness

Total number of items in the stomachs of Sacramento River juvenile chinook salmon was variable depending on fish size and month of capture (Table 4). Due to prey size differences, the SFI provides greater insight into fish energy intake than does total number of items consumed (Hyslop 1980). The mean SFI of juvenile salmon from the Sacramento River was 2.4% of wet weight, comparable to other reports for juvenile chinook salmon feeding in the wild (Sagar and Glova 1988, Brodeur 1992). Brodeur (1992) estimated a geometric mean stomach fullness of 1.34% ($n = 734$) for juvenile chinook salmon off the coast of Washington and Oregon; his range of estimates (0–8%) was very similar to the range in our study (0.5–8.4%). Sagar and Glova (1988) found mean percent dry weight of food per fish varied between 1.3% and 3.4% for juvenile chinook salmon over one 24-hour period in the Rakaia River, New Zealand. Our estimate does not, however, reflect maximum feeding intensity, as most samples were taken in the morning and early afternoon. Results from a 24-hour study of drift and diet in the Sacramento River indicated maximum

feeding occurred near dusk (Petrusso⁹ 1998). Twenty-four-hour consumption by juvenile chinook salmon in that study was estimated as approximately 12.5% of wet body weight, comparable to the 8.3% of dry weight estimated by Sagar and Glova (1988), and corresponding to roughly half the maximum consumption for juvenile salmonids (about 17–20%) (Beauchamp et al. 1989). The estimate reflects reasonably good feeding under natural conditions and is corroborated by condition of juvenile chinook salmon (Petrusso⁹ 1998).

CONCLUSIONS

The numeric abundance and taxonomic composition of the food resources in the Sacramento River were comparable to other river systems (Becker 1973; Sagar and Glova 1987, 1988; Bowles and Short 1988). Juvenile chinook salmon appeared to have access to abundant foods, although the largest, most preferred sizes of invertebrates were not always available. As most fall-run fry leave the upper river and arrive at the Sacramento-San Joaquin Delta by April during wet years, food availability during the study period may have increased due to seasonal changes in environmental conditions (increased temperature and water clarity) as well as a reduction in juvenile salmon density (intraspecific competition). Thus, the juvenile salmon that remained during April–June 1996 may have benefited from both factors, and the observed stomach fullness of salmon in the upper river may reflect these favorable conditions. An interesting direction for future studies could be to determine the food availability and energy intake of fall-run juveniles in the upper river during dry years, when emigration is delayed.

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LITERATURE CITED

- Arnett, R.H. Jr. 1985. American insects: A handbook of the insects of America north of Mexico. Van Nostrand Reinhold, New York, New York, USA.

⁹Petrusso, P.A. 1998. Feeding habits and condition of juvenile chinook salmon in the upper Sacramento River, California. M.S. Thesis, Michigan State University, East Lansing, Michigan, USA.

- Beauchamp, D.A., D.J. Stewart, and G.L. Thomas. 1989. Corroboration of a bioenergetics model for sockeye salmon. *Transactions of the American Fisheries Society* 118:597-607.
- Becker, C.D. 1970. Feeding bionomics of juvenile chinook salmon in the central Columbia River. *Northwest Science* 44:75-81.
- Becker, C.D. 1973. Food and growth parameters of juvenile chinook salmon, *Oncorhynchus tshawytscha*, in the central Columbia River. *Fishery Bulletin* 71:387-400.
- Bowles, D.E. and R.A. Short. 1988. Size composition of invertebrate drift and fish predation in a Texas stream. *The Southwestern Naturalist* 33:177-184.
- Brodeur, R.D. 1992. Factors related to variability in feeding intensity of juvenile coho salmon and chinook salmon. *Transactions of the American Fisheries Society* 121:104-114.
- Brown, R.L. 1991. Bioengineering problems in river systems of the Central Valley, California. *American Fisheries Society Symposium* 10:19-31.
- Cada, G.F., J.M. Loar, and M.J. Sale. 1987. Evidence of food limitation of rainbow and brown trout in southern Appalachian soft-water streams. *Transactions of the American Fisheries Society* 116:692-702.
- Field-Dodgson, M.S. 1988. Size characteristics and diet of emergent chinook salmon in a small, stable, New Zealand stream. *Journal of Fish Biology* 32:27-40.
- Fisher, F.W. 1994. Past and present status of Central Valley chinook salmon. *Conservation Biology* 8:870-873.
- Healey, M.C. 1983. Coastwide distribution and ocean migration patterns of stream- and ocean-type chinook salmon, *Oncorhynchus tshawytscha*. *Canadian Field-Naturalist* 97:427-433.
- Higgs, D.A., J.S. Macdonald, C.D. Levings, and B.S. Dosanjh. 1995. Nutrition and feeding habits in relation to life history stage. Pages 161-315 in: C. Groot, L. Margolis, and W.C. Clarke, editors. *Physiological ecology of Pacific salmon*. UBC Press, Vancouver, British Columbia, Canada.
- Hyslop, E.J. 1980. Stomach contents analysis - a review of methods and their application. *Journal of Fish Biology* 17:411-429.
- Keeley, E.R. and J.W.A. Grant. 1997. Allometry of diet selectivity in juvenile Atlantic salmon (*Salmo salar*). *Canadian Journal of Fisheries and Aquatic Science* 54:1894-1902.
- Merritt, R.W. and K.W. Cummins, editors. 1984. *An introduction to the aquatic insects of North America*, second edition. Kendall/Hunt Publishing Co., Dubuque, Iowa, USA.
- Merz, J.E. and C.D. Vanicek. 1996. Comparative feeding habits of juvenile chinook salmon, steelhead, and Sacramento squawfish in the lower American River, California. *California Fish and Game* 82:149-159.
- Pennak, R.W. 1978. *Freshwater invertebrates of the United States*, second edition. John Wiley & Sons, New York, New York, USA.
- Rondorf, D.W., G.A. Gray, and R.B. Fairley. 1990. Feeding ecology of subyearling chinook salmon in riverine and reservoir habitats of the Columbia River. *Transactions of the American Fisheries Society* 119:16-24.
- Sagar, P.M. and G. J. Glova. 1987. Prey preferences of a riverine population of juvenile chinook salmon, *Oncorhynchus tshawytscha*. *Journal of Fish Biology* 31:661-673.
- Sagar, P.M. and G.J. Glova. 1988. Diel feeding periodicity, daily ration and prey selection of a riverine population of juvenile chinook salmon, *Oncorhynchus tshawytscha* (Walbaum). *Journal of Fish Biology* 33:643-653.
- Strauss, R.E. 1979. Reliability estimates for Ivlev's electivity index, the forage ratio, and a proposed linear index of food selection. *Transactions of the American Fisheries Society* 108:344-352.

Usinger, R.L. 1956. Aquatic insects of California. University of California Press, Berkeley, California, USA.

Wainwright, P.C. and B.A. Richard. 1995. Predicting patterns of prey use from morphology of fishes. *Environmental Biology of Fishes* 44:97-113.

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Appendix A. Relative abundance of items in the Sacramento River drift (n = 107 drift samples) and diet of juvenile chinook salmon (n = 153 fish) for the period April–June 1996.

Taxon	Life stage	Drift		Diet	
		Number	Percent	Number	Percent
INSECTA					
Collembola	All stages	112	1.08	2	0.04
Ephemeroptera					
Baetidae	Adult	132	1.27	57	1.02
Caenidae	Adult	20	0.19	–	–
Unidentified	Adult	7	0.07	31	0.56
Baetidae	Nymph	1,103	10.61	750	13.44
Heptageniidae	Nymph	10	0.10	6	0.11
Tricorythidae	Nymph	10	0.10	9	0.16
Unidentified	Nymph	8	0.08	11	0.20
Other ^a	All stages	18	0.17	4	0.07
Total Ephemeroptera	All stages	1,308	12.58	868	15.56
Odonata	Nymph	1	0.01	–	–
Orthoptera	Nymph	1	0.01	–	–
Dermaptera	Adult	1	0.01	–	–
Plecoptera					
Perlodidae	Nymph	4	0.04	9	0.16
Unidentified	Nymph	3	0.03	7	0.13
Other ^a	All stages	11	0.11	7	0.13
Total Plecoptera	All stages	18	0.17	23	0.41
Psocoptera	Adult	24	0.23	–	–
Hemiptera					
Corixidae	Nymph	26	0.25	5	0.09
Other ^a	All stages	24	0.23	10	0.18
Total Hemiptera	All stages	50	0.48	15	0.27
Homoptera					
Aphididae	Adult	71	0.68	135	2.42
Cicadellidae	Adult	25	0.24	21	0.38
Coccoidea	Adult	18	0.17	–	–

Appendix 1. (Cont'd)

Taxon	Life Stage	Drift		Diet	
		Number	Percent	Number	Percent
Unidentified	Adult	1	0.01	19	0.34
Unidentified	Nymph	13	0.13	6	0.11
Other ^a	All stages	10	0.10	3	0.05
Total Homoptera	All stages	138	1.33	184	3.30
Thysanoptera	All stages	8	0.08	1	0.02
Coleoptera					
Staphylinidae	Adult	24	0.23	9	0.16
Unidentified	Adult	12	0.12	4	0.07
Dytiscidae	Larva	13	0.13	5	0.09
Other ^a	All stages	25	0.24	6	0.11
Total Coleoptera	All stages	74	0.71	24	0.43
Diptera					
Cecidomyiidae	Adult	23	0.22	-	-
Chironomidae	Adult	1,189	11.43	1,587	28.44
Empididae	Adult	2	0.02	39	0.70
Ephydriidae	Adult	10	0.10	23	0.41
Simuliidae	Adult	24	0.23	32	0.57
Unidentified	Adult	27	0.26	18	0.32
Chironomidae	Pupa	228	2.19	786	14.09
Simuliidae	Pupa	-	-	12	0.22
Chironomidae	Larva	4,199	40.38	1,175	21.06
Muscidae	Larva	-	-	6	0.11
Simuliidae	Larva	433	4.16	121	2.17
Other ^a	All stages	25	0.24	22	0.39
Total Diptera	All stages	6,160	59.24	3,821	68.48
Trichoptera					
Hydropsychidae	Adult	5	0.05	23	0.41
Hydroptilidae	Adult	7	0.07	8	0.14
Psychomyiidae	Adult	18	0.17	3	0.05
Unidentified	Adult	1	0.01	25	0.45
Brachycentridae	Larva	11	0.11	-	-
Hydropsychidae	Larva	47	0.45	124	2.22
Glossosomatidae	Larva	17	0.16	24	0.43
Psychomyiidae	Larva	11	0.11	1	0.02
Unidentified	Larva	21	0.20	3	0.05
Other ^a	All stages	10	0.10	8	0.14
Total Trichoptera	All stages	148	1.42	219	3.92
Lepidoptera	All stages	4	0.04	-	-
Hymenoptera					
Formicidae	Adult	19	0.18	15	0.27
Unidentified	Adult	73	0.70	73	1.31
Other ^a	All stages	14	0.13	2	0.04
Total Hymenoptera	All stages	106	1.02	90	1.61
TOTAL INSECTA		8,153	78.40	5,247	94.03

Appendix 1. (Cont'd)

Appendix 1: (Cont'd)

Taxon	Life Stage	Drift		Diet	
		Number	Percent	Number	Percent
MISCELLANEOUS					
Hydroida		90	0.87	1	0.02
Nematoda		55	0.53	64	1.15
Oligochaeta		595	5.72	9	0.16
Hydracarina		82	0.79	4	0.07
Arachnida		33	0.32	22	0.39
Cladocera					
Bosminidae		11	0.11	—	—
Daphniidae		5	0.05	—	—
Macrothricidae		340	3.27	—	—
Unidentified		142	1.37	6	0.11
Total Cladocera		498	4.79	6	0.11
Eucopepoda					
Calanoida		118	1.13	—	—
Cyclopoida		399	3.84	—	—
Harpacticoida		1	0.01	—	—
Total Eucopepoda		518	4.98	—	—
Teleostei	Larva	352	3.38	147	2.63
Catostomidae	Larva	4	0.04	37	0.66
Total Teleostei		356	3.42	184	3.30
Other ^a		13	0.13	7	0.13
TOTAL MISCELLANEOUS		2,240	21.54	297	5.32
UNIDENTIFIED	All stages	6	0.06	36	0.65
Terrestrial Subtotal		409	3.93	353	6.33
Aquatic Subtotal		9,990	96.07	5,227	93.67
GRAND TOTAL		10,399	100.00	5,580	100.00

^aCombines taxa making up <0.1% of the drift or diet.

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